

Sprout

Autonomous Greenhouse Plant-Care System

ROB-GY 6103 (Advanced Mechatronics) · NYU Tandon · Group 15 · Raj Priyadarshi, Devank Thawre, Sagarika Rao Valluri

Abstract

Sprout is an autonomous mechatronic system that automates the care of greenhouse plants through a coordinated multi-bot architecture driven by real-time soil sensor data. It handles watering, fertilisation, and plant-health monitoring across a 3×2 instrumented plant bed with no manual intervention.

Three specialised bots share the plant bed and coordinate wirelessly over the ESP-NOW protocol: a water bot, a compost/NPK bot, and a Raspberry Pi camera bot. Splitting the work keeps each bot mechanically simple and lets the tasks run independently and in parallel.

Technical Highlights

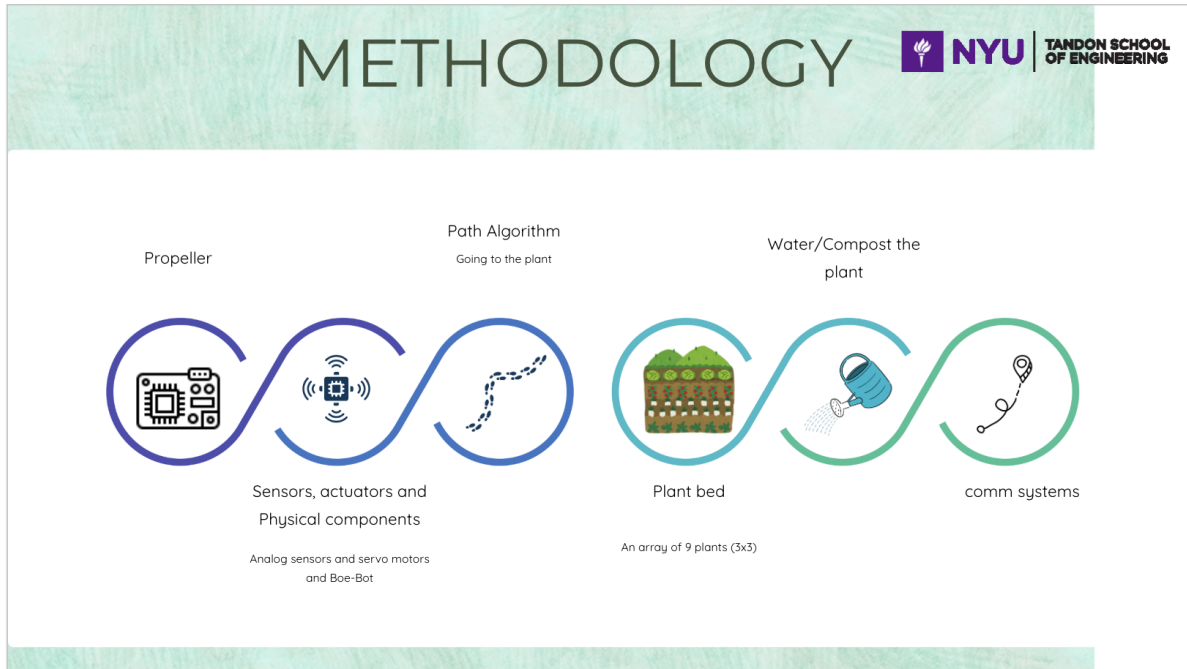
- **Precision watering:** capacitive soil-moisture sensors drive a solenoid valve to dispense a calibrated dose (2 ml per pulse, 1–6 pulses) per plant, tuned from gravimetric soil profiles.
- **Autonomous navigation:** IR line-following plus IR object detection let each bot drive the track and stop precisely at every plant.
- **Soil-nutrient sensing:** an NPK sensor over RS-485 Modbus RTU measures nitrogen, phosphorus, and potassium; a servo arm lowers the probe and a compost flap dispenses accordingly.
- **Wireless coordination:** ESP-NOW peer-to-peer links between ESP32 modules pass per-plant readings with no Wi-Fi router.
- **Remote monitoring:** a Raspberry Pi camera bot serves plant images through a Flask web server with AI plant identification.

Technologies Used

- **Languages:** C++ (Arduino), Python (Flask)
- **Controllers:** Arduino UNO, ESP32 (ESP-NOW), Raspberry Pi
- **Sensors:** Capacitive soil moisture, NPK (RS-485 Modbus RTU), IR line/object, Pi Camera

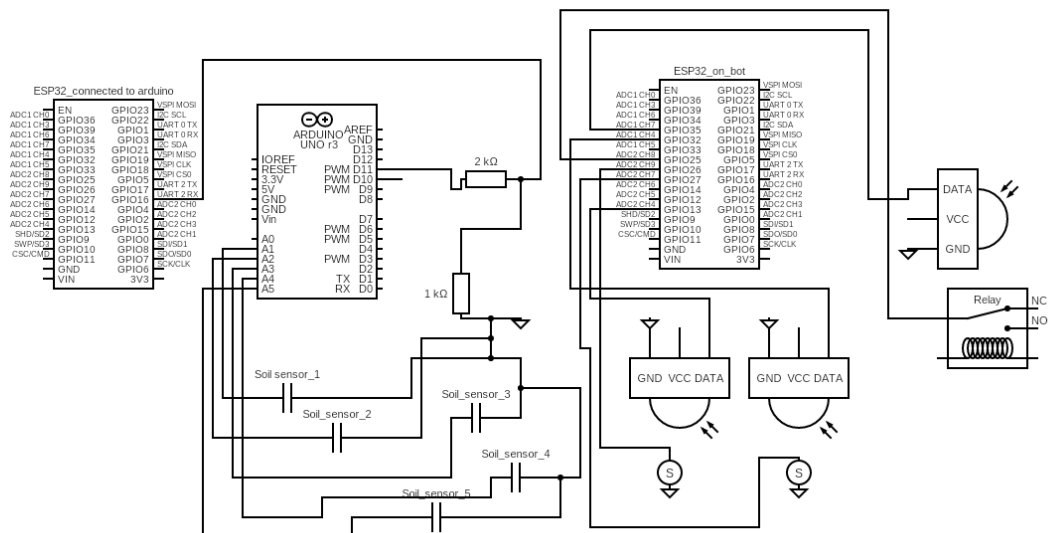
- **Actuators:** Solenoid valve, servo-driven compost flap, drive motors
- **Protocols:** UART/Serial, ESP-NOW (Wi-Fi peer-to-peer), RS-485 Modbus
- **Web/AI:** Flask web server with AI plant identification

Figures

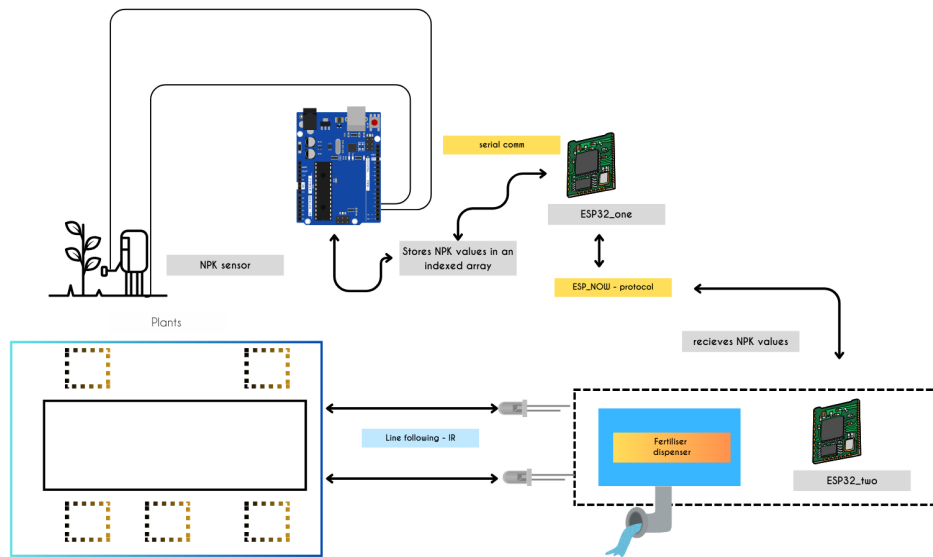


System methodology across drive, sensing, path planning, and dispensing.

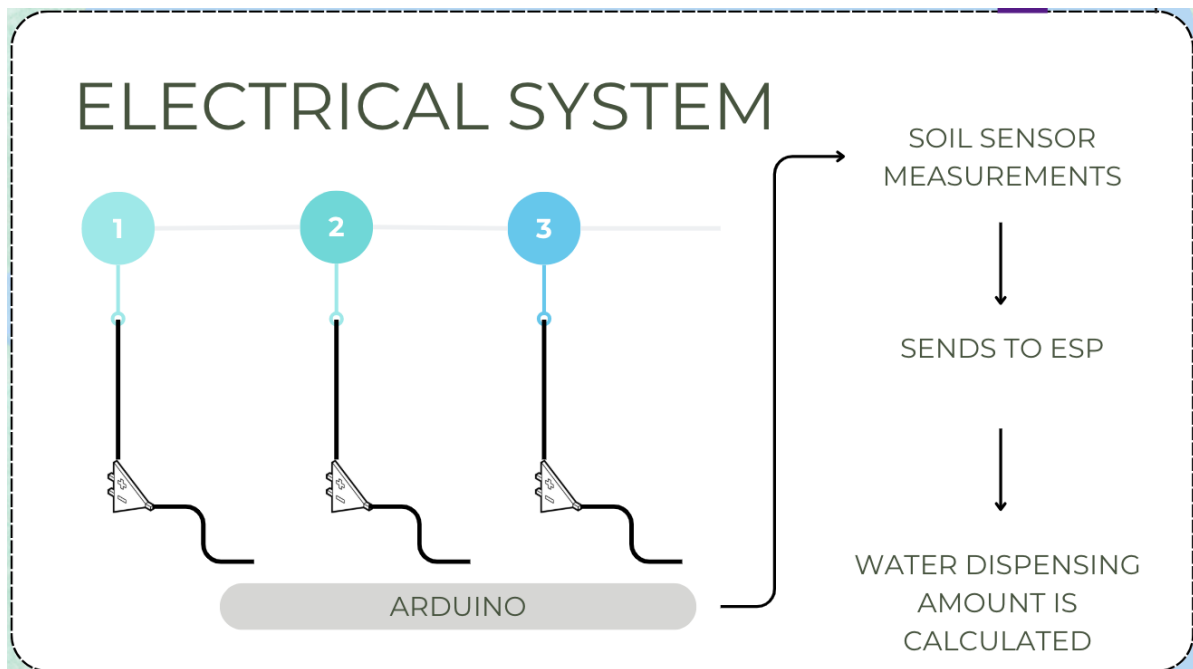
Soil_sensor_5



Full circuit schematic — Arduino UNO, two ESP32s, soil sensors, NPK sensor, relay, and servos.



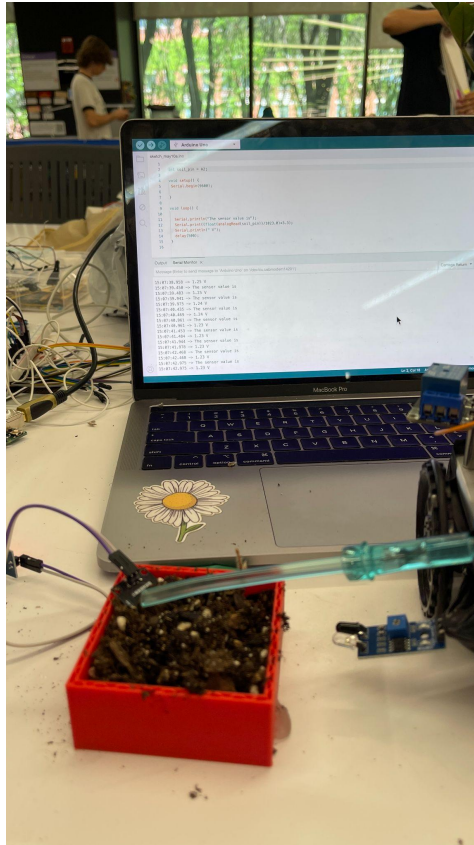
System block diagram of the sensing chain and ESP-NOW links.



Electrical-system flow from soil measurement to computed water dose.



Calibrating dispense count to soil-moisture voltage.



Two soil profiles characterised via gravimetric/volumetric analysis.

Outcome

Sprout demonstrates fully autonomous, sensor-driven multi-bot coordination: soil-moisture voltages map cleanly to water doses, NPK readings guide fertilisation, and a camera bot adds remote plant-health monitoring — all wirelessly orchestrated over ESP-NOW.